VOLATILE OILS

Topic Outline

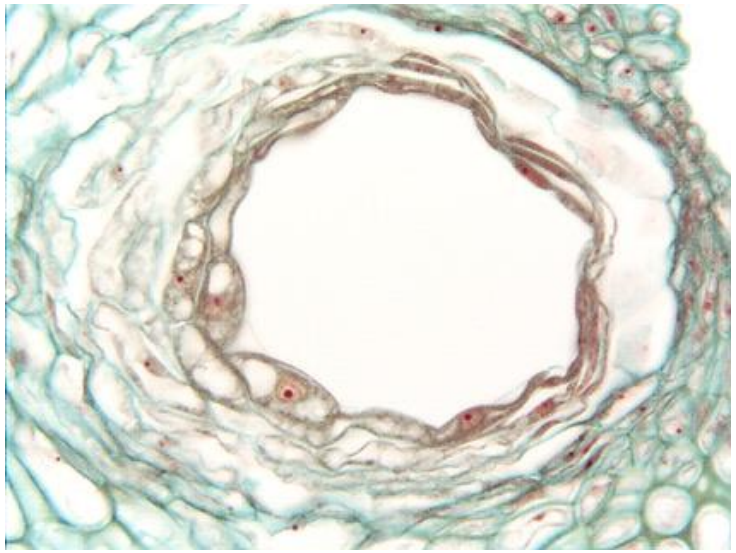
- Distribution and occurrence
- Uses
- Methods of preparation
  - Distillation
  - Expression
  - Extraction with solvent
  - Enzymatic hydrolysis
- Physical properties
- Chemistry
- Biosynthesis
- Constitution of volatile oils
  - Hydrocarbons, Alcohols, Aldehydes, Ketones, Esters, Phenols and phenolic ethers, Oxides, Peroxides, Sulfur containing compounds Nitrogen containing compound

Definition

- Defined as oily liquids, which are entirely, or almost entirely, volatile without decomposition
- Volatile or essential oils, as their name implies, are volatile in nature.
- The term terpene represents the hydrocarbons (C5H8)n, while terpenoids include the hydrocarbons, as well as their oxygenated derivatives.
- They are generally mixture of terpene hydrocarbons (C_{10} H_{22}) and oxygenated terpene compounds derived from these hydrocarbons
  - Oil of turpentine-mainly HC's
  - Oil of Clove – mainly oxygenated compounds

Introduction

- Terpenes, or terpenoids, are the largest group of secondary products (metabolites).
- They are all formed from acetyl CoA or glycolytic intermediates.
- All official volatile oils are of vegetable origin.
- Normally pre-exist in the plant – stored in a special secretory tissue (e.g. Citrus peel oil cells or oil ducts in umbelliferous fruits).
  - EXCEPTION: Oil of bitter almond – formed by hydrolysis of the glycosides.



Volatile oils Vs Fixed Oils

- Volatile oils are different from fixed oils in that they:
  - Are not glyceryl esters of long chain fatty acid (they are hydrocarbons)
  - Do not leave permanent grease spot on paper
  - Can not be saponified with alkalis
  - Do not become rancid like fixed oils but will oxidize and resinfy up on exposure to light(O_2)

Distribution

- Widely distributed in plant Kingdom (more than 2000spp. belonging to about 90 families)
  - Main families include: *Pinaceae, Lauraceae, Rutaceae, Myrtaceaae, Umbelliferae, Labiatae and Compositae*
- Very few volatile oils occur in animals e.g. Skatol from glandular secretion of Civet cat.

Occurrence

- The odorous, volatile principles of plants and animal sources are known as volatile oils.
- All the official volatile oils are of vegetable origin.
- In most instances the volatile oil pre-exists in the plant.
- In some instances the volatile oil does not pre-exist, but is formed by the decomposition of a glycoside.
- They are frequently associated with other substances such as gums and resins

Occurrence

- Synthesis & accumulation of essential oils are generally associated with the presence of specialized histological structures, often located on or near the surface of the plant:
  - Oil cells of Zingiberaceae
  - Oil tubes(ducts) of Umbuliferea
  - Glandular trichomes of Lamiaceae
  - Secretory cavities of Myrtaceae or Rutaceae
  - Secretory canals of Apiaceae or Astereraceae (Compositeae)
  - Internal lysigenous or schizogenous passangers of pinaceae, and rutaceae

Why volatile oils are present in plants?

- Waste products: No biological role
- Detoxification products: elimination of metabolic products
- Attraction of insects aiding in pollination
- Insect repellents defence agents

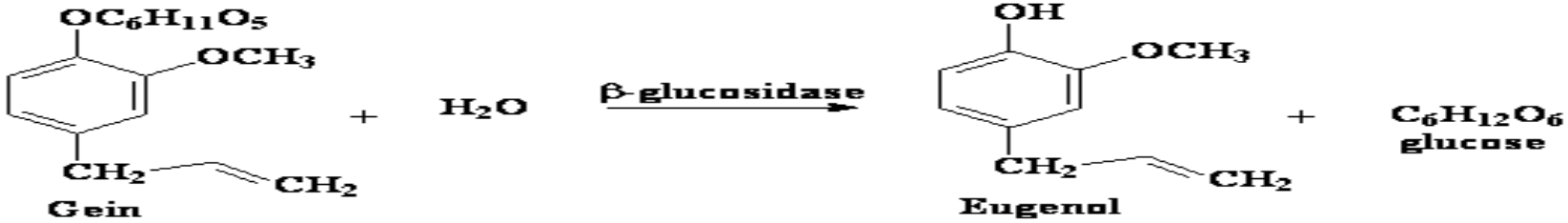
USES OF VOLATILE OILS

- **Therapeutically** (Oil of Eucalyptus)
- **Flavouring** (Oil of Lemon)
- **Perfumery** (Oil of Rose)
- **Starting materials** to synthesize other compounds (Oil of Turpentine)
- **Anti-septic** – due to high phenols (Oil of Thyme). Also as a preservative (oils interfere with bacterial respiration)
- **Anti-spasmodic** (Ginger, Lemon balm, Rosemary, Peppermint, Chamomile, Fennel, Caraway)
- **Aromatherapy**

1- Methyl Salicylate (Wintewr green oil)



2- Eugenol



3- Allyisothiocyanate



4- Benzaldehyde



5- Glucovanillin



Physical properties of volatile oils

- Although volatile oils differs greatly in their chemical constitution, they have a number of physical properties in common
  - Posses characteristic odour
  - Characterized by high refractive indices
  - Most of them are optically active
  - Their density is lower than water
    - Except sassafras oil, clove oil and cinnamon oil
  - As rule volatile oils are water immiscible with water, but they are sufficiently soluble to impart their odour to water

PROPERTIES OF VOLATILE OILS

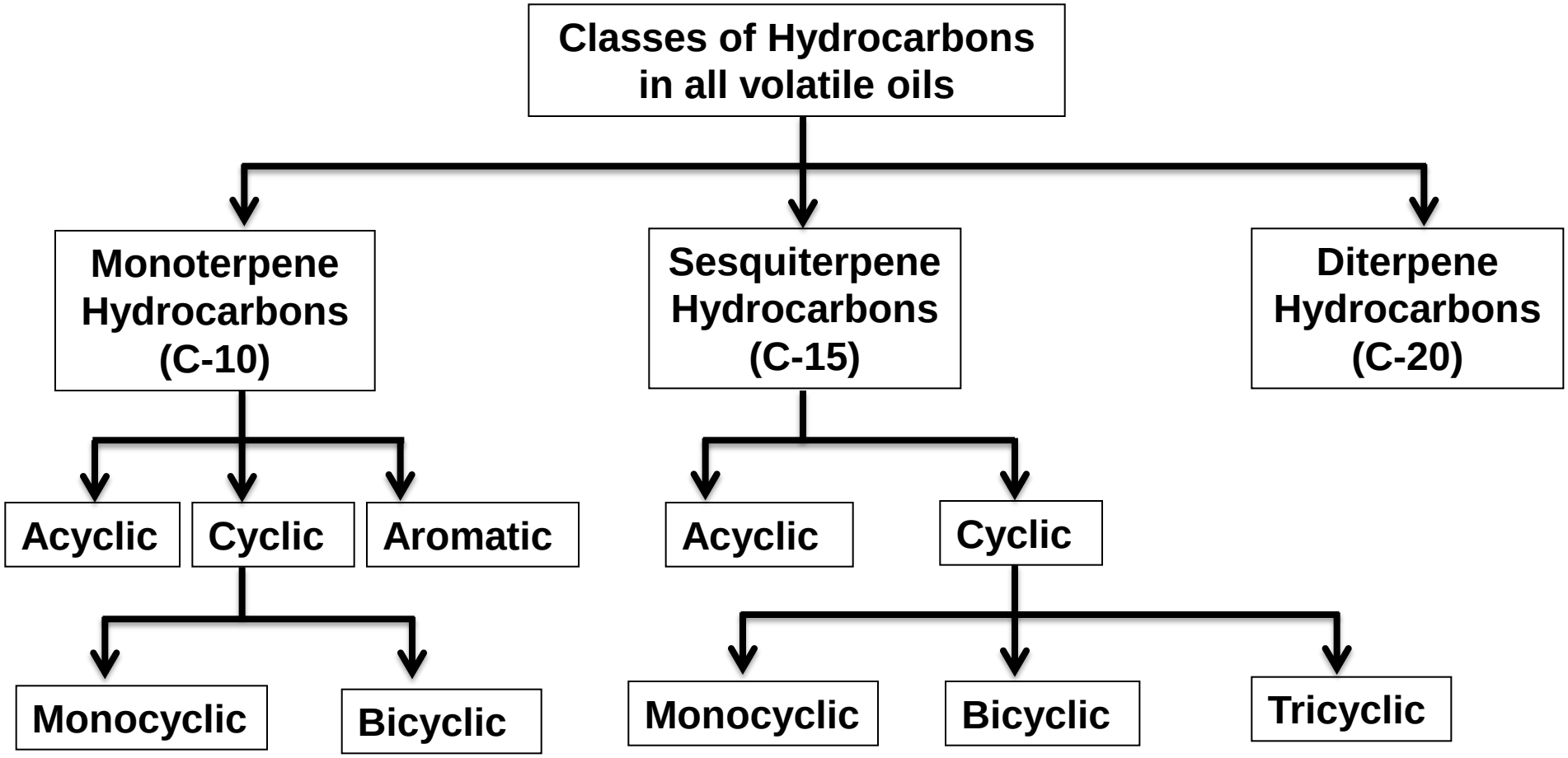
- Almost **entirely volatile without decomposition**.
- Density: Most are **less than 1g/ml**.
  - 2 oils are heavier – Oil of **Cinnamon** and **Clove**.
- Soluble in ether, chloroform & alcohol.
- Slightly soluble in water: give it a **characteristic odour & taste.**
- Leaves a temporary translucent stain on paper which disappears as the oil volatilizes.
- Most are colourless. Oxidize on exposure to air and resinify → colour becomes **darker** (odour changes slightly).

Chemistry of Volatile Oils

- Essential oils chemistry is very complex in nature as essential oils themselves have many chemical ingredients.
- Some play a major part and others a minor part.
- The ingredients found in essential oils are organic due to their molecular structure which is based on carbon atoms held together by hydrogen atoms.
- Oxygen atoms and sometimes nitrogen and sulphur atoms are also present.

Constituents of Volatile Oils

- **Hydrocarbons** Occur practically in all volatile oils & can be classified as follows

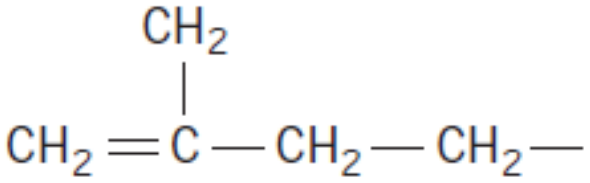


- Below is a list of the main chemical components found in essential oils:
  - **Monoterpenes**
  - **Sesquiterpenes**
  - **Phenols**
  - **Alcohols**
  - **Ethers / Esters**
  - **Ketones**
  - **Coumarins**
  - **Hazardous essential oils**

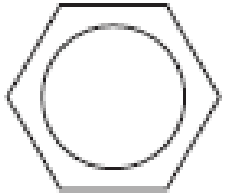
Constitution of volatile oils

- All essential oils are made up of carbon, oxygen and hydrogen atoms; the chemistry of essential oils is complex but, in simple terms, all essential oils hold a combination of molecules made up of the following chemical components:
  - Terpene hydrocarbons
    - Monoterpene hydrocarbons
    - Sesquiterpenes

- Oxygenated compounds
  - Phenols
  - Alcohols
    - Monoterpene alcohols
    - Sesquiterpene alcohols
  - Aldehydes
  - Ketones
  - Esters
  - Lactones
  - Coumarins
  - Ethers
  - Oxides
- Phenols and phenolic ethers
- Oxides, Peroxides, Sulfur containing compounds
- Nitrogen containing compound



Isoprene unit



Aromatic ring

Properties of essential oils

Compound	Properties
Hydrocarbons	Stimulant, decongestant, antiviral, antitumour
Alcohols	Antimicrobial, antiseptic, tonifying, spasmolytic
Sesquiterpene alcohols	Anti-inflammatory, anti-allergenic
Phenols	Antimicrobial, irritant, immune stimulating
Aldehydes	Spasmolytic, sedative, antiviral
Cyclic aldehydes	Spasmolytic,
Ketones	Mucolytic, cell-regenerating, neurotoxic
Esters	Spasmolytic, sedative, antifungal
Oxides	Expectorant, stimulant
Coumarins	UV sensitising, antimicrobial
Sesquiterpenes	Anti-inflammatory, antiviral
Phenylpropanes	Carminative, anaesthetic
Sesquiterpene Lactones	Mucolytic, immune stimulating

Hydrocarbon Volatile Oils

- *Hydrocarbons occur practically in all volatile oils*.
- Hydrocarbon can be classified biogenetically into:
  - Monoterpene Hydrocarbons (C10)
  - Sesquiterpene Hydrocarbons (C15)
  - Diterpene Hydrocarbons (C20)

- **Hydrocarbons** reduce inflammation, bacterial infections and intestinal gas.
- They also tend to be mentally stimulating.
- Hydrocarbons are one of the most common types of chemical found in essential oil.
- Some of the aromatherapy essential oils that contain this chemical compound are: cypress, ginger, nutmeg, patchouli, sandalwood, pine, fir, lemon and carrot seed.

➢ Terpenes Hydrocarbons

Monoterpene hydrocarbons

Sesquiterpenes

✓ Monoterpene hydrocarbons

- These monoterpene compounds are found in nearly all essential oils and have a structure of 10 carbon atoms and at least one double bond. The 10 carbon atoms are derived from two isoprene units.
- They react readily to air and heat sources.
- For this reason citrus oils do not last well, since they are high in monoterpene hydrocarbons and have a quick reaction to air, and are readily oxidized.

- Although some quarters may simply state that these components have anti-inflammatory, antiseptic, antiviral and antibacterial therapeutic properties while some can be analgesic or stimulating with a tonic effect, it could be seen as a very broad generalization, since this large group of chemicals vary greatly.
- Since some have a stimulating effect on the mucus membranes they are also often used as decongestants.
- Many monoterpenes are antimicrobial due to which they protect the plants.

- Monoterpene hydrocarbons may be classified into:
  - **Acyclic** [*aliphatic, not cyclic*]
    - Myrcene
    - Ocimene
  - **Aromatic**
    - *P-Cymene*
  - **Alicyclic**
    - *Monocyclic.*
      - ***P-Menthane***
    - *Bicyclic.*
      - **α-Pinene**

LEMON PEEL

- **Biological source:**
  - Lemon peel is the outer part of the pericarp of the ripe or nearly ripe fruits of *Citurs limonis* Linn.
- **Family:** Rutaceae
- **Geographical source:**
- It is indigenous to India, but also cultivated in countries like Spain, Italy and Sicily.
- **Description:**
  - Colour - (i) Fresh lemon peels outer surface is bright greenish yellow, internally white.
    (ii) Dried peels outer surface is yellow and inner surface pithy white.
  - Odour - Strong, fragrant, aromatic and characteristic
  - Taste - Aromatic, bitter.

- **Chemical constituents:**
  - Lemon peel contains 2 to4% volatile oil.
  - The volatile oil contains mainly limonene (about 90%), citral(about 4%) and other aromatic compounds like geranyl acetate and terpineol.
  - The other constituents of the peel are Hesperidin (Flavonoid), pectin, Mucilage and Calcium Oxalate.
- **USES:**
  - It is used as carminative and stimulant and oil is used as a perfuming and flavouring agent.
  - It is used for extraction of pectin and volatile oil.

Camphor oil

- Biological source:
  - Camphor oil is obtained from the wood by steam distillation of *Cinnanomum camphora*
- Family: Lauraceae.
- Geographical source:
  - Indigenous to Eastern Asia. Abundant in Taiwan, China and Japan.
- Description:

Colour	-	Colourless or yellowish liquid
Odour	-	Camphorous
Solubility	-	Soluble in chloroform, alcohol and ether.

Camphor oil

- Chemical constituents:
  - It mainly contains saffrole, acetaldehyde, dipentene, camphor eugenol, d-pinene, eucalyptol, phellandrene and cineole.
- Uses:
  - Camphor oil is used as rubefacient, counter irritant and as flavour in soaps and tooth powder and cosmetic products.
  - It is used for isolation of camphor.

NUTMEG & NUTMEG OIL

- **Biological source:**
  - Nutmeg is the dried kernel of the seed of *Myristica fragrans*
- **Family:** Myristicaceae

➢ Geographical Source:

- Indigenous to the Molucca Islands (Spice Islands)
- Cultivated in Indonesia, Malaysia, India, West Indies and other tropical countries.

➢ Description:

Colour	-	Externally the kernels are greenish brown or brown
Odour	-	Strongly aromatic
Taste	-	Pungent and aromatic

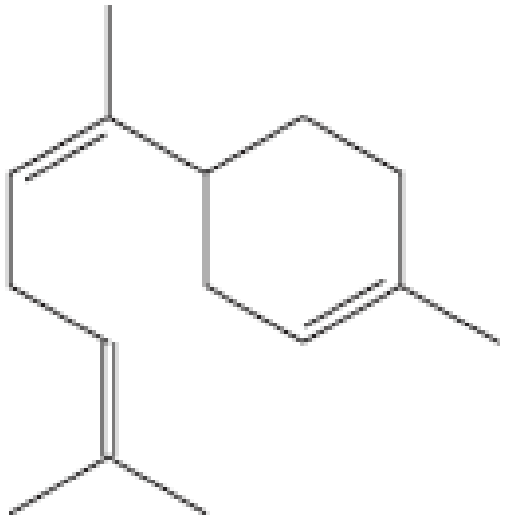
- **Chemical constituents:**
- Nutmeg contains 5 to 16% volatile oil.
  - The volatile oil contains about 4 to 8% myristicin, elimicin and saffrole.
  - Myristicin – a benzene: toxic to humans (large does of nutmeg or nutmeg oil may cause convulsions).
- Contains about 30% of fat. The fat of nutmeg is also known as nutmeg butter.
- geraniol, terpineol, d-Camphene, Dipentene, Eugenol & eugenol derivatives
- The other constituents of the drug are protein and starch.
- **Uses:**
  - It is used as an aromatic, a stimulant and carminative.
  - It is also used as a flavouring agent.
  - The fat is used in soap industries.
  - The fat and volatile oil is used in the treatment of rheumatism.

- Sesquiterpenes
  - These sesquiterpenes consist of 15 carbon atoms and have complex pharmacological actions and here we can look at chamazulene, which is found in German chamomile.
  - It has anti-inflammatory and anti-allergy properties. Another sesquiterpene often found in chamomile and rose, as well as other floral oils is farnesene.

SESQUITERPENES ($C_{15}H_{24}$)

- These essential oils contain in the fractions having boiling point between 250-280°C
- They have complex pharmacological actions.
- They have anti-inflammatory and anti-allergy properties.
- Being unsaturated hydrocarbons, the sesquiterpenes form, hydrogen halides and nitorosylchlorides,.
- Some of them are crystalline.

- ***Acyclic*** e.g. Franesene, Sesquicitronellene.
- ***Monocyclic*** e.g. Zingiberene,
- ***Bicyclic*** e.g. Cadinene, Azulene and Guiazulene.
- ***Tricyclic*** e.g. α-santalene.



Bisabolene

GINGER

- **Biological source:**
  - Ginger is the peeled or unpeeled rhizome of *Zingiber officinale*
- **Family:** Zingiberaceae
- **Geographical source:**
  - It is said to be native of South East asia, but is cultivated in carribean islands, Africa, Australia, mauritius, jamaica, Taiwan and India.
- Description:
  - Colour - Externally it is buff coloured
  - Odour - Agreeable and aromatic
  - Taste - agreeable and pungent

- **Chemical constituents**:

- Ginger consists of Volatile oils (1 – 4%)
  - Camphene, Cineole, Citral, Borneol
- Gingerol – pungent component – Anti-inflammatory
- Sesquiterpene HC's
  - Zingiberene & Zingiberol (Sesquiterpene alcohol)
- Resinous matter(5-8%), Starch (40-60%), Fat(10%), Fibre(5%).

- **Uses**:
  - It is used as a stomachic, an aromatic, a carminative, stimulant, Anti-emetic, Anti-bacterial & Anti-fungal and flavouring agent.
  - Ginger oil is used in mouth washes, beverages and liquors.
  - Ginger powder is effective in motion sickness.
  - Gingerol – pungent component – Anti-inflammatory
  - Shogaols– increases bile secretion & enhanced GIT activity.

OXYGENATED TERPENE HYDROCARBONS

- **General characters:**
  - include (**alcohols, esters, ethers, aldehydes, ketones, phenols, oxides, peroxides and** etc.).
  - They have the advantage of being the *principal odour carriers*
  - Oxygenated terpene compounds have better solubility in dilute alcohol.
  - have greater stability against oxidising and resinifying influences than hydrocarbons.
  - Terpeneless (free from HCS) volatile oils are more soluble in alcohol and more stable against oxidation and resinification.

- **Isolates or Aromatic isolates:** "Are the odourful compounds isolated from certain essential oil
  - **Citral** isolated from lemon grass oil
  - **Eugenol** isolated from Clove oil
  - **Citronellal** isolated from Citronella oil

- **Oxygenated mono terpene hydrocarbons**
  - Terpene alcohols
  - Terpene phenols and phenolic ethers
  - Terpene aldehyde
  - Terpene ketones
  - Terpene oxides
  - Terpene peroxide
  - Terpene esters
  - Sulphur and Nitrogen containing compounds

- **Alcohols** act as diuretics; they kill bacteria, energize and stimulate the body.
- Tea tree, ginger, rose, rosewood, peppermint, sandalwood and patchouli are some of the plants that contain alcohols.

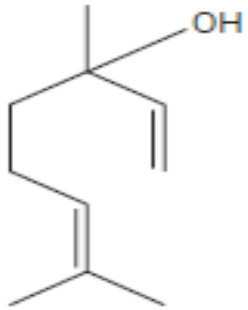
➢ Terpene alcohols can be

- ***monoterpene alcohols***
- ***sesquiterpene alcohols***

Alcohols

- Monoterpene alcohols
  - These oils have good antiseptic, anti-viral and anti-fungal properties with very few side effects such as skin irritation or toxicity and have an uplifting energizing effect.
  - Examples of these alcohols are linalool, citronellol and terpineol found respectively in lavender, rose and geranium, and in juniper and tea tree oil.

- **Linalool** can help relieve discomfort. It may be found in rosewood and lavender.



- **Citronellol** may be found in rose, lemon, eucalyptus, geranium, and others.
- **Geraniol** may be found in geranium as well as palmarosa.
- **Farnesol** may be found in chamommile. It is also good for the mucous.
- Other terpene alcohols include borneol, menthol, nerol, terpineol, vetiverol, benzyl alcohol, and cedrol.

Lavender oil

- **Biological source**:
  - It is the volatile oil obtained by steam distillation of fresh flowering tops of *Lavandula officinalis* Chaix
- **Family**: Labiatae.
- **Description**:

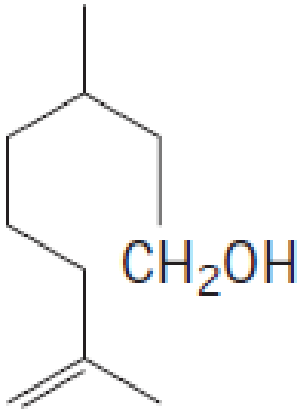
Colour	-	Colourless
Odour	-	Characteristic with pleasant aroma
Taste	-	Aromatic, pungent
Solubility	-	slightly soluble in water, soluble in alcohol, CS_2.

- **Chemical constituents**:
  - The oil contains about 30-40% esters, calculated as linalyl acetate and also linalool, pinene, geraniol and cineol.
- **Uses**:
  - It is used as aromatic
  - As carminative
  - As flavouring agent in perfumery and cosmetics.

- Sesquiterpene alcohols
  - These alcohols are not commonly found in essential oils, but when found, like bisabolol in German chamomile, have great properties, which include liver and glandular stimulant, anti-allergen and anti-inflammatory.
  - Other oils that contain sesquiterpene alcohols are sandalwood (a-santalol) as well as ginger, patchouli, vetiver, carrot seed, everlasting and valerian.
    - **Bisabolol** is one of the strongest sesquiterpene alcohols. It may be found in chamomile oils where it also functions well as a fixative.

Classification monoterpene alcohols

1. **Acyclic (aliphatic)**
2. **Cyclic (alicyclic)**



Citronellol

1. Acyclic (aliphatic)

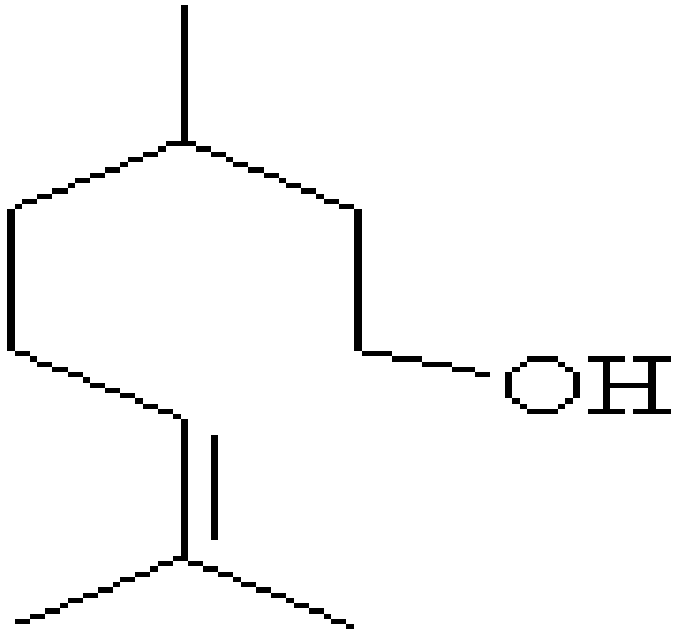
- **Citronellol**
- **Geraniol**
- **Nerol**
- **Linalol (linalool)**

2. Cyclic (alicyclic)

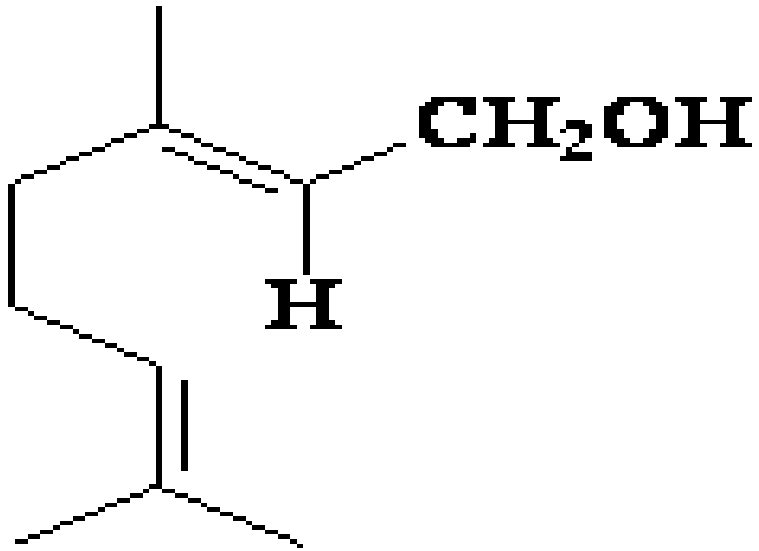
- **Monocyclic**
  - *menthol*
  - *α-terpineol*
- **Bicyclic**
  - *Borneol*
  - *Isoborneol*
- **Tricyclic**
  - *Santalol*

- **1. Acyclic (aliphatic) monoterpene alcohol:**

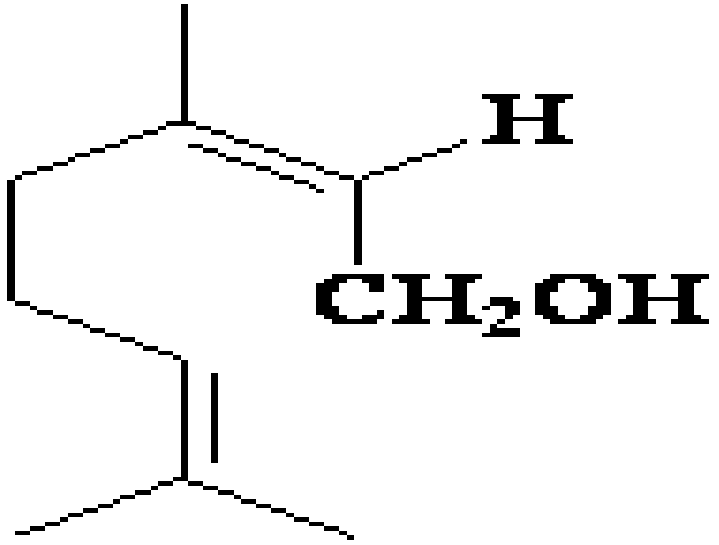
    1) *Citronellol*



2) Geraniol



3) Nerol

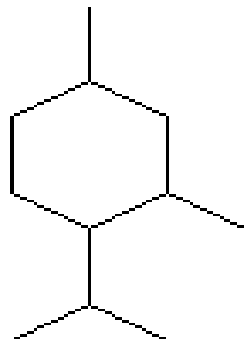


- H-CH_3 opposite side (*trans*) H-CH_3 in the same side (*cis*)

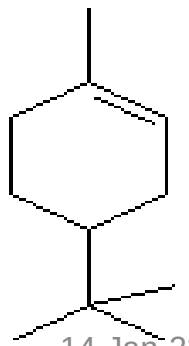
- **Cyclic (alicyclic) *monoterpene alcohols***

2. Cyclic (Alicyclic) alcohols

A. Monocyclic

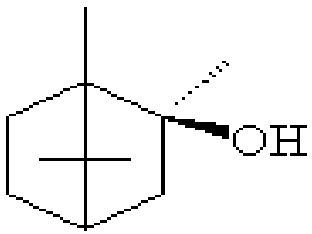


Menthol

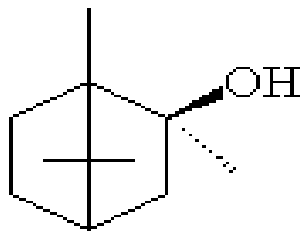


α-terpineol

B. Bicyclic

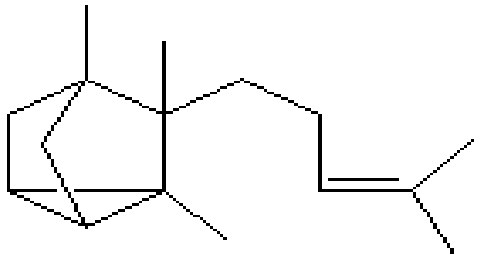


Borneol



Isoborneol

C. Sesquiterpene alco.



Santalol

to which the medicinal value of sandal wood is due

Oil of citronella

- **Biological source**:
  - It is obtained by steam distillation from the fresh leaves of *Cymbopogan nardus* Linn.
- **Family**: Graminae
- **Geographical source**:
  - It is indigenous to Srilanka and cultivated in Myanmar, Malaysia, Indonesia and India.
- **Description**

Colour	-	Pale greenish yellow
Odour	-	aromatic
Taste	-	pungent

- **Chemical constituents**:
  - Citronella grass contains 0.3 to 0.9% of strongly aromatic and pleasantly flavoured volatile oil.
  - It mainly contains geraniol(40-60%) and citronellal(5-20%).
  - Other constituents of the oil are d-camphene, limonene, dipentene, borneol, and methyl eugenol.
- **Uses**:
  - It is used as perfume for soaps, brilliantines.
  - In mosquito repellent creams and sprays.
  - As a flavouring agent for liniments and lotions.

Oil of sandalwood

- **Biological source**:
  - It is obtained from the heart-wood of *Santalum album* Linn
- **Family**: Santalaceae
- **Geographical source**:
  - These plants are found in India and malaysia
- **Description**:

Colour	-	Pale yellow to colourless
Odour	-	Characteristic, persistant
Taste	-	Unpleasant
Solubility	-	slightly soluble in water and soluble in alcohol and chloroform

- **Chemical constituents**:
  - Oil of sandal wood contains about 95% of isomeric sesquiterpene alcohols, α – santalol and β – santalol.
  - Additionally, the oil contains an aldehyde santalal, santene, santanone, teresantol, santalone and santalene.
- **Uses**:
  - The oil is used for symptomatic treatment of dysuria and in diminishing the frequency of micturation marked in the tuberculosis of bladder.
  - It is mainly used as a perfume in preparation of cosmetics and incense sticks.

CORIANDER

- **Biological source:**
  - These are fully dried ripe fruits of plant known as *Coriandrum sativum* Linn.
- **Family:** Umbelliferae
- **Geographical source:**
  - Indigenous to Italy.
  - Also cultivated in Holland, Central & Eastern Europe, Mediterranean (Morocco, Malta & Egypt), China and India.
- **Description:**
  - Colour - Yellowish brown to brown
  - Odour - Aromatic
  - Taste - spicy and characteristic

- **Chemical constituents**:
  - 0.3 - 1.8 % Volatile oils
  - Volatile oil of drug contains about 90% of **D-Linalool/coriandrol and coriandryl acetate**.
  - Small quantities of L-borneol, geraniol and Pinene.
  - Coriander leaves are rich in Vitamin A content.
  - Non-linalool alchols & esters
  - Flavonoids, Coumarins, Phenolic acids,High fat content (16-28%), Protein (11-17%),
- **Uses**:
  - The fruits as well as volatile oil are used as an aromatic, carminative, stimulant and flavourign agent.
  - Coriander oil is used alongwith purgatives to prevent gripping.
  - It is an ingredient of compound spirit of orange and cascara elixir.
  - Domestic purposes (cooking - curries)

Terpene Phenols and Phenolic Ethers

- Phenols are similar to alcohols in essential oils but are stronger in their actions; confusingly, phenol names also end in *-ol* so it is essential to identify the individual chemical component and know if it is an alcohol or a phenol.
- Phenol properties include being antiseptic, bactericidal and a stimulant; however, phenols can be an irritant if overused in large quantities.

Terpene Phenols and Phenolic Ethers

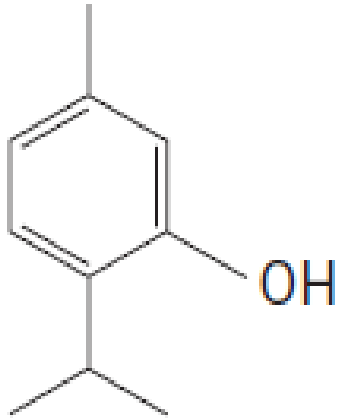
- **General characters**
  - Phenols form water-soluble phenate salts with alkali such KOH and NaOH.
  - They can be separated from other contaminants, using the anion exchange resins.
  - They are aromatic compounds; thus, undergo nitration, sulfonation and halogenation.
  - They give blue colour with $FeCl_3$ and phosphomolybdic acid (phenolic hydroxyls).
  - **Phenols** are another common chemical found in most aromatherapy oils.

Terpene Phenols and Phenolic Ethers

- Since they contain both the flavour and disinfectant properties, this makes them an excellent compound for oral and hygienic preparations.
- Phenols are powerful antibacterials.
- They warm the skin and stimulate blood flow.
- Because of their heating action, oils containing phenols should be used in small amounts.
- Basil, clove, thyme and oregano oils contain phenols.

- **Phenols** - are responsible for the fragrance of an oil.
  - They are antiseptic, anti-bacterial, and strongly stimulating but can also be quite caustic to the skin.
  - They contain high levels of oxygenating molecules and have anioxidant properties.
  - **Eugenol** may be found in clove and cinnamon oil.
  - **Thymol** is found in thyme and may not be as caustic as other phenols.
  - **Carvacrol** may be found in oregano and savory. It is estimated that it may possibly contain some anti-cancerous properties.
  - Others in the phenol family include methyl eugenol, methyl chavicol anethole, safrole, myristicin, and apiol.

- **Classification**
  - ***Phenolic volatile oils***
  - ***Monohydric phenols***
    1. Carvacrol
    2. Eugenol
    3. Thymol
  - ***Dihydric phenols***
    1. ***Catechol***



Thymol

CLOVE



- **Biological source:**
  - Cloves consists of dried flower buds of *Eugenia caryophyllus*
- **Family:** Myrtaceae.

➢ **Geographical Sources:**

➢ It is indigenous to Molucca islands. Also cultivated in Zanzibar, pemba, penang, madagascar, carribean Islands, Indonesia, srilanka, India & Brazil.

➢ **Description:**

Colour	-	Crimson to dark brown.
Odour	-	slightly aromatic
Taste	-	Pungent and aromatic followed by numbness

- **Chemical constituents**:
  - Clove contains about 14-21% Volatile oils mainly eugenol(about 70-90%) & isoeugenol, eugenol acetate, Sesquiterpenes (α and β caryophyllenes)
  - Small quantities of esters, ketones and alcohols.
  - Tannins(10-13%).
  - Resin
- **Uses**:
  - Clove is used as dental analgesic, carminative, stimulant, flavouring agent, an aromatic and antiseptic.
  - It is also used in the preparation of cigarettes.
  - The oil is used in perfumery and also in the manufacture of vanillin.

- **Phenolic ethers**

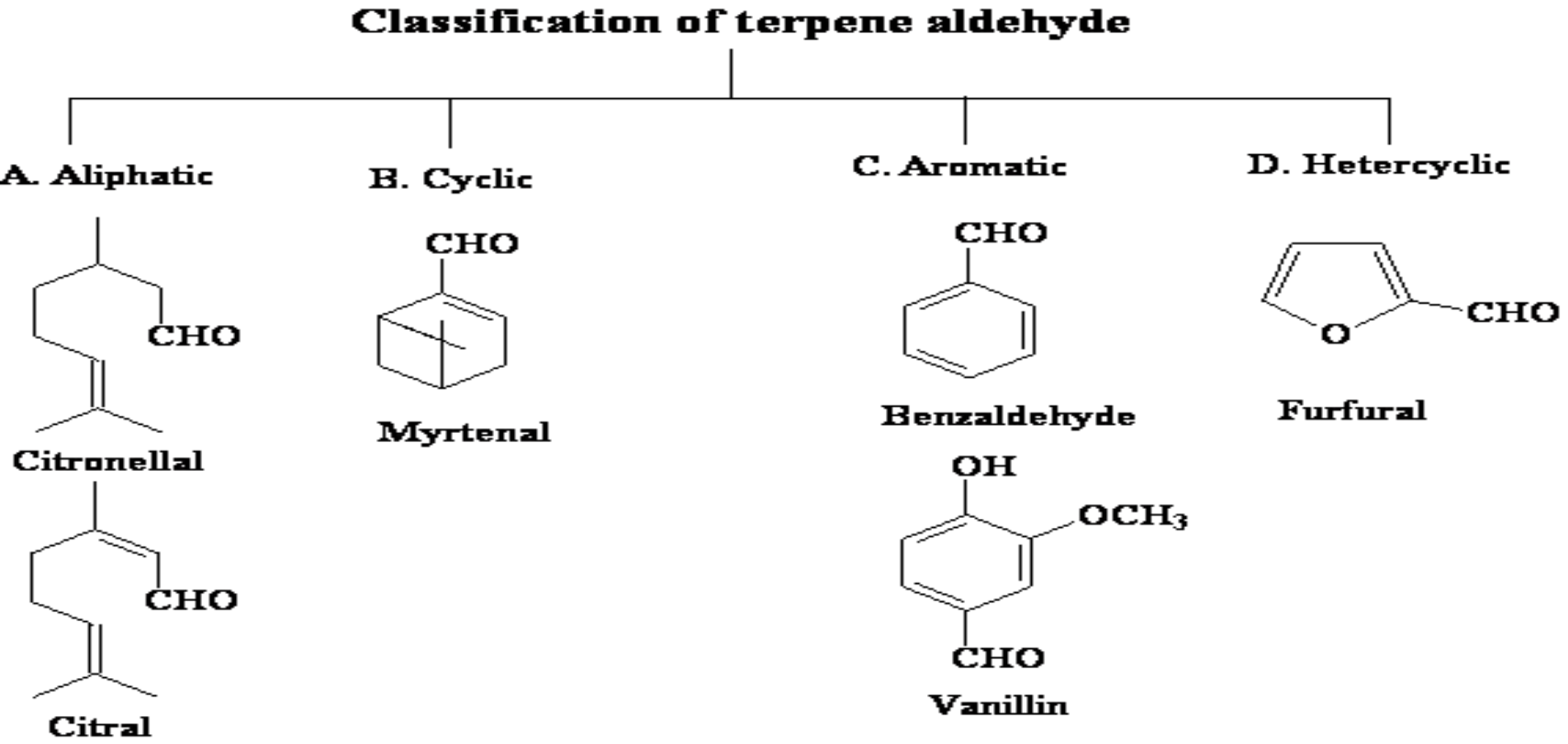
1. **Anethol**
2. **safrole**
3. **myristicin**
4. **apoile**

Aldehydes

- These aldehydes have anti-fungal, anti-inflammatory, disinfectant, sedative yet uplifting therapeutic qualities and are the component that imparts the citrus-like fragrance in melissa, lemongrass and citronella.
- These properties are best used in aromatherapy when the essential oil is used in low dilutions - around 1%.
- Should oils high in this component be used, it could cause skin irritation and sensitivity as for instance lemongrass oil.
- Aldehydes are also unstable and will easily oxidize in the presence of oxygen and even low heat.

Terpene Aldehydes

Classification of ***Terpene Aldehydes***



- Citral is used as starting materials for the synthesis of β-ionone which is a precursor of vitamin A.

Lemon-grass oil

- **Biological source**:
  - The lemon grass oil is a volatile oil obtained by steam distillation from the leaves and aerial parts of the plants *Cymbopogan flexuousus*
- **Family**: Graminae
- **Geographical source**:
  - It is indigenous to India and found wild as well as cultivated throughout India. It occurs in west africa, East africa and Guatamela.
- **Description**:

Colour	-	Yellowish brown liquid
Odour	-	Resembling to lemon oil
Taste	-	similar to lemon oil

- **Chemical constituents**:
  - Lemon grass oil chiefly contains citral in addition to methylheptenol, nerol, citronellal, dipentene and geraniol.
- **Uses**:
  - It is used as a flavouring agent and In perfumery.
  - It is employed as a source of citral from which β - ionine is prepared which is a starting material for the synthesis of vitamin A.

CINNAMON

- **Biological source:**
  - Cinnamon consists of the dried bark of the shoots of coppiced trees of *Cinnamomum zeylanicum*
- **Family:** Lauraceae
- **Geographical source:**
  - Cultivated in Sri Lanka, South India. Also cultivated in Seychelles, Madagascar, Martinique, Jamaica & Brazil.
- **Description:**
  - Colour - The outer surface is dull yellowish brown, while the inner surface is dark yellowish brown
  - Odour - fragrant
  - Taste - aromatic and sweet followed by ward sensation.
  - Shape - found in the form of compound quills.

- **Chemical constituents:**
  - 0.5 to 1.0% Volatile oils
  - Cinnamon oil contains 60-70% of cinnamaldehyde, 5-10% of eugenol, benzaldehyde, cuminaldehyde and other terpenes like phellandrene, pinene, cymene, caryophyllene, etc.
  - The other constituents are 1.2% of tannins(Phlobaphenes), mucilage, Calcium Oxalate, Starch and a sweet substance called mannitol.

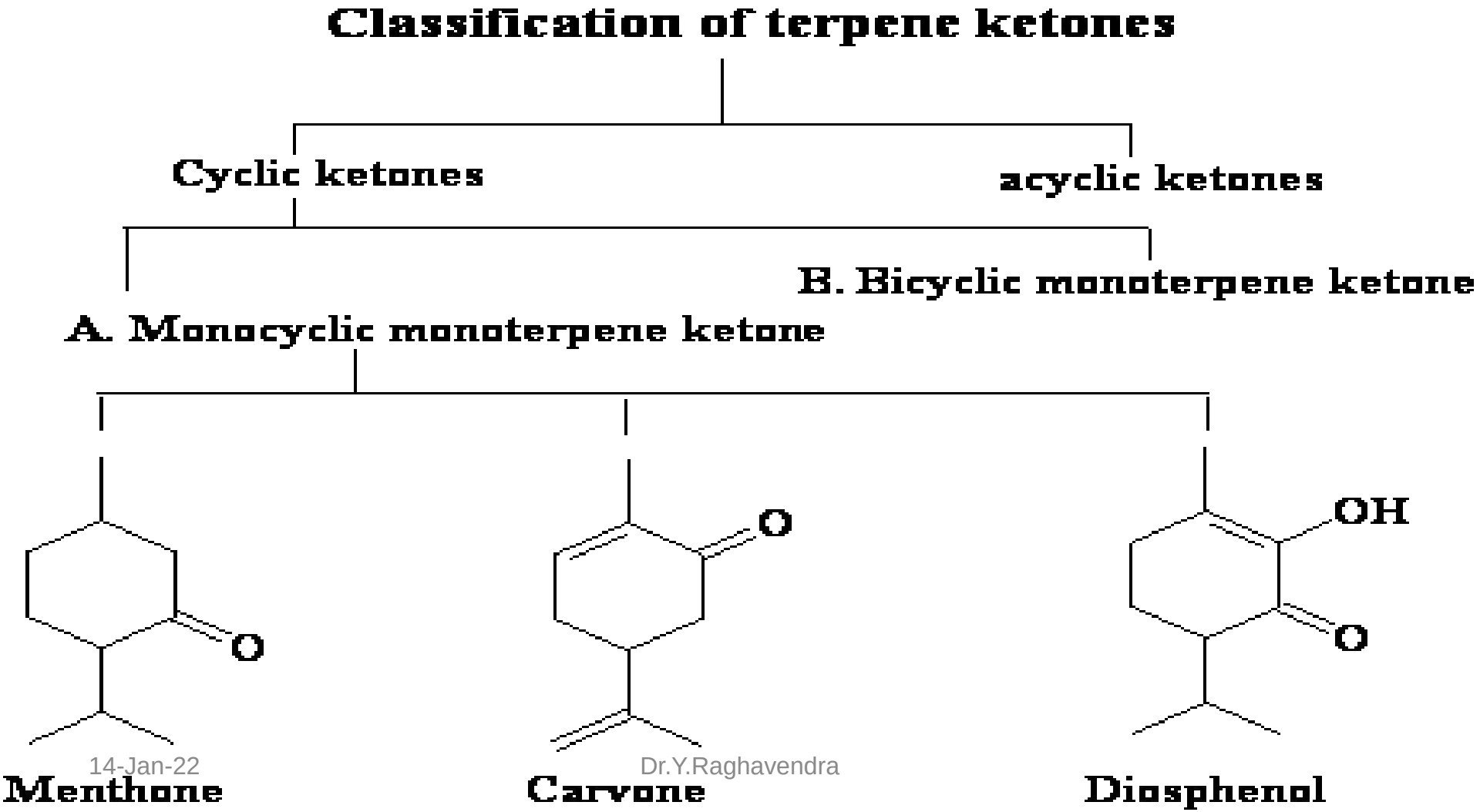
- **Uses:**
  - The bark is used as a carminative, stomachic and mild astringent.
  - It is also used as a flavouring agent, stimulant, an aromatic and antiseptic.
  - Commercially it is used as a spice and condiment and also in preparation of candies, dentrifices and perfumes.

Ketones

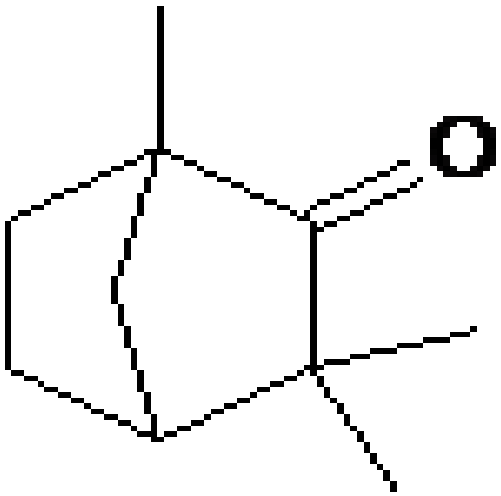
- Although ketones can be toxic, as in the case of thujone found in thuja and wormwood oil as well as pinocamphone found in others, they also have some great therapeutic benefits - especially in the field of easing the secretion of mucus as well as cell and tissue regeneration.
- The ketone italidone found in everlasting, not only has the mucolytic (mucus easing) properties, but is also useful in skin regeneration, wound healing and reducing old scar tissue such as in wounds, stretch marks and adhesions.
- Essential oils high in ketones need to be used with care in pregnancy.
- **Ketones** promote new cell growth and help wounds heal faster and they're found in rosemary, camphor, sage and eucalyptus.

Terpene Ketones

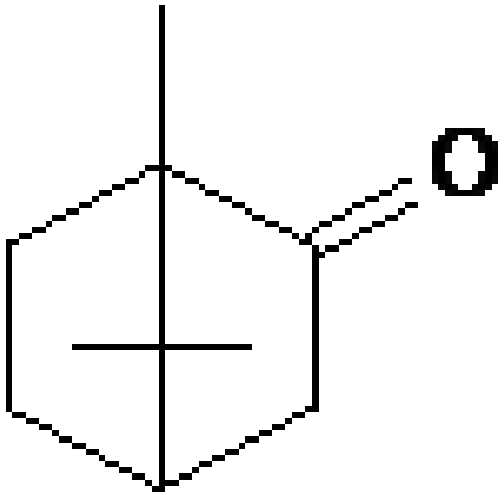
- Bicyclic monoterpine ketone
- **Fenchone ,Camphor**



Fenchone



Camphor



FENNEL



- **Biological source:**
  - Fennel consists of the dried ripe fruits of the plant known as *Foeniculum vulgare* Miller
- **Family:** Umbelliferae

➢ **Geographical source:**

- It is indigenous to meditteranean countries and largely cultivated in Romania, Russia, Germany, France, India, Japan, China & Egypt.

➢ **Description:**

Colour	-	Green to yellowish brown
Odour	-	sweet aromatic
Taste	-	strongly aromatic

- **Chemical constituents**:
  - Fennel consists of 3-7 % Volatile oil.
  - The chief active constituent of volatile oil is a ketone, Fenchone(about 20%, colourless pungent liquid with aromatic odour) and
  - Phenolic ether anethole(about 50%, sweet in taste and odour).
  - The other constituents are phellandrene, limonene, methyl chavicol, anisic aldehyde etc.
- **Uses**:
  - It is used as carmiantive, aromatic and stimulant.
  - It is also an expectorant.
  - Pharmaceutically it is used as flavouring agent.

CARAWAY & CARAWAY OIL

- **Biological source:**
  - Caraway consists of the dried ripe fruits of *Caram carvi Linn.*
- **Family:** Umbelliferae
- **Geographical source:**
  - Indigenous to Holland Central & Northern Europe.
  - It is also cultivated in Central Asia and other european countries like Denmark, Germany, Finland, Poland, Hungary, Britain and also in India, Egypt, Morocco, Australia & China.
- **Description**:

Colour - Yellowish brown

Odour - characteristic and aromatic

Taste - characteristic and aromatic

- **Chemical constituents**:
  - Caraway consists of 2-8 % Volatile oils.
  - Volatile oil contains 45-65% of Carvone, Limonene, dihydro carvone and traces of carvacrol.
  - About 10 % Fixed oils, 15% Proteins and resins.
- **Uses**:
  - It is used as an aromatic, a stimulant and carminative.
  - It is used as a spice for culinary purposes and also for flavouring the bread, biscuits, cheese and cakes.

Aromatics Oils That Have a Significant % of Ketones

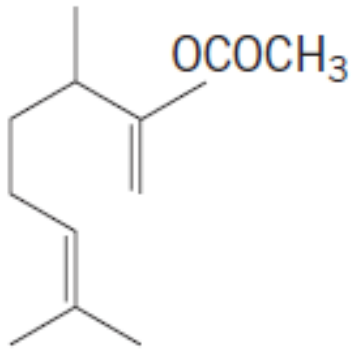
- **Eucalyptus Dives** - *Eucalyptus dives*
- **Finger root** - *Boesenbergia pandurata*
- **Peppermint** - *Mentha x piperita*
- **Rosemary ct camphor** - *Rosmarinus officinalis ct camphor/ 1,8 cineole*
- **Rosemary ct verbenone** - *Rosmarinus officinalis ct verbenone/camphor*
- **Sitka Spruce** - *Picea sitchensis*
- **Spike Lavender** - *Lavandula latifolia*
- **Turmeric** - *Curcuma longa*
- **Vetiver** - *Vetiveria zizanoides*
- **White Spruce** - *Picea glauca*

Esters

- Esters are formed from alcohols and acids, and are named after both their original molecules with the alcohols dropping the "ol" and gaining an "yl" and the acids dropping the "ic" and gaining an "ate".
- The esters found in essential oils are normally very fragrant and tend to be fruity and their therapeutic effects include being sedative and antispasmodic.
- Some esters also have anti-fungal and anti-microbial properties - like the anti-fungal properties in geranium oil.
- many volatile oils owe their odour or perfume and flavouring properties due to ester.

Terpene esters

- The esters in essential oils are normally very fragrant and tend to be fruity and their therapeutic effects include being sedatives, and anti spasmodic.
- The most well known ester must be linalyl acetate, which is found in lavender, clary sage as well as petitgrain.
  - **Benzyl Benzoate**
  - **Methyl salicylate**
  - **Methyl anthranylate**



linalyl acetate

PEPPERMINT OIL

- **Biological source:**
  - The oil is obtained by steam distillation of the fresh flowering tops of the plant known as *Mentha piperita Linn.*
- Family: Labiatae.
- **Geographical source:**
  - USA, Japan, England, France, Italy, Russia and India.
- Description:
  - Colour - Colourless to yellow
  - Odour - Characteristic and pleasant
  - Taste - Pungent followed by cooling sensation.
  - Solubility - Soluble in 70% alcohol, ether, chloroform.

- **Chemical constituents**:
  - It should contain at least 1.2 % volatile oil.
  - Oil contain chiefly *l* – menthol to an extent of 70 % in the form of esters.
  - 15-32 & menthone and 4.5-10 % menthyl acetate.
  - The other terpenes include

    Menthone

    Isomenthone

    Menthyl acetate

    Limonene

    Cineole

    Menthofuran,

    Jasmone

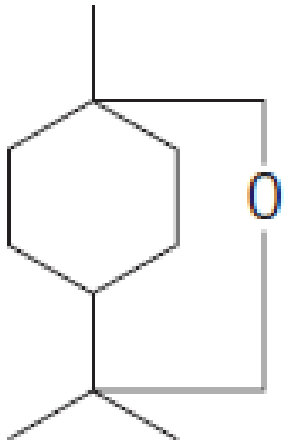
    Camphene etc.

Oxides

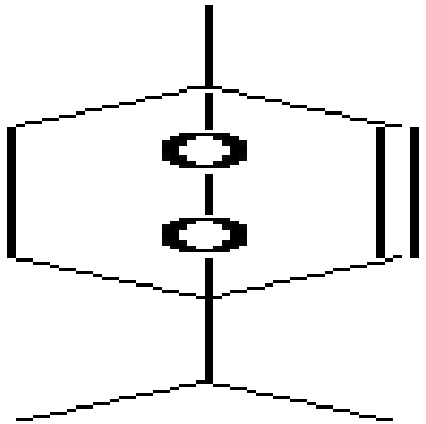
- The main therapeutic effect of oxides are that of expectorant, with 1,8-cineole - commonly known as eucalyptol being the most well known.
  - Cineol (or eucalyptol) is by far the most important member of the family and virtually exists in a class of its own. It is anesthetic, antiseptic, and works as an expectorant. Cineol is well known as the principal constituent of eucalyptus oil. It may also be found in rosemary, cinnamon, melissa, basil, and ravensara.
  - Other oxides include linalol oxide, ascaridol, bisabolol oxide, and bisabolone oxide.

Terpene Oxide and Peroxide

1,8 Cineol (Eucalyptol)



Ascaridol



Eucalyptus oil

- **Biological source:**
  - Eucalyptus oil is the volatile oil obtained by distillation of fresh leaves of *Eucalyptus globulus*.
- **Family**: Myrtaceae.
- **Geographical source:**
  - Indigenous to Australia. Also cultivated in Portugal, SA, Spain, China, Brazil and India.
- **Description:**
  - Colour - Colourless to pale yellow liquid
  - Odour - Aromatic
  - Taste - pungent and camphorous followed by cold.
  - Solubility- Soluble in 90%alcohol, fats,fixed oil and in paraffin.

- **Chemical constituents**:
  - Volatile Oil (at least 2 %)
  - Chiefly contains cineole, also known as eucalyptol(about 80%).
  - It also contains pinene, camphene and traces of phellandrene, citronellal and geranyl acetate.
- **Uses**:
  - It is used as counter-irritant, an antiseptic and expectorant.
  - It is used to relieve cough and in chronic bronchitis in the form of inhalations.
  - It is an ingredient of several liniments and ointments.
  - Solution of eucalyptus oil is used as nasal drops.
  - sesquiterpene - Anti-bacterial action against oral pathogens.

CARDAMOM



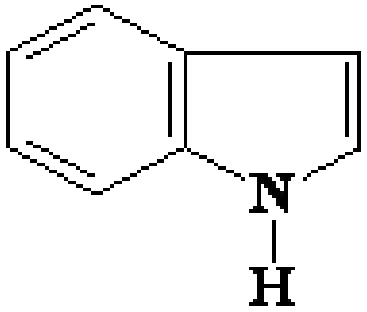
- **Biological source:**
  - Cardamom consists of the dried, nearly dried ripe fruits of *Elettaria cardamomum* var *minuscula*
- **Family:** Zingiberaceae.
- **Geographical source:**
  - Guatemela is the largest producer in the world. Also cultivated in India and Srilanka.
- **Description:**

Colour	-	Natural colour is green or pale buff and after treated with sulphurdioxide) it changes to white.
Odour	-	Aromatic, agreeable and pleasant
Taste	-	Strongly aromatic

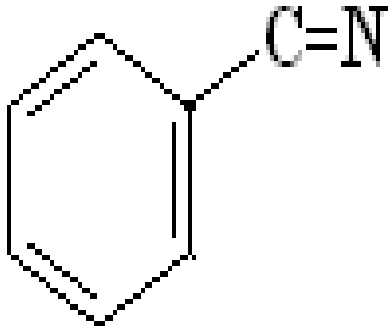
- **Chemical constituents**:
  - 2 – 8 % volatile oil.
  - The active constituent of the VO is cineole.
  - Other aromatic compounds present are terpinyl acetate, terpineol, borneol, terpinene etc.
  - Abundant starch (up to 50%) Fixed oil (1 – 10%), protein and Calcium oxalate.
- **Uses**:
  - It is used as an aromatic, as carminative and as stimulant.
  - It is also a good flavouring agent in preparation of curries, biscuits, sweets etc.
  - It is used in the form of compound tincture.

Terpene Sulphur and nitrogen containing compounds

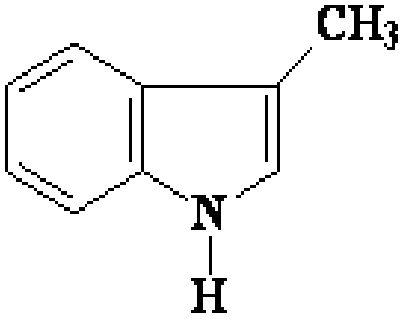
Indol



Benzyl cyanide



Skatol



Civet

- **Biological source**:
  - Civet is the odourous secretion obtained from highly specialised scent glands in the region of the external generative organs of male and female civet cat, *Viverra zibetha* Linn.
- **Family**: Viverridae
- **Description**:

Colour	-	Pale yellow semi-liquid
Odour	-	obnoxious odour, when diluted becomes very pleasant.
Taste	-	bitter
Solubility	-	Partly soluble in hot alcohol & ether.

- Civet cats are found all over South East Asia, Africa, Madagascar and Southern Europe.

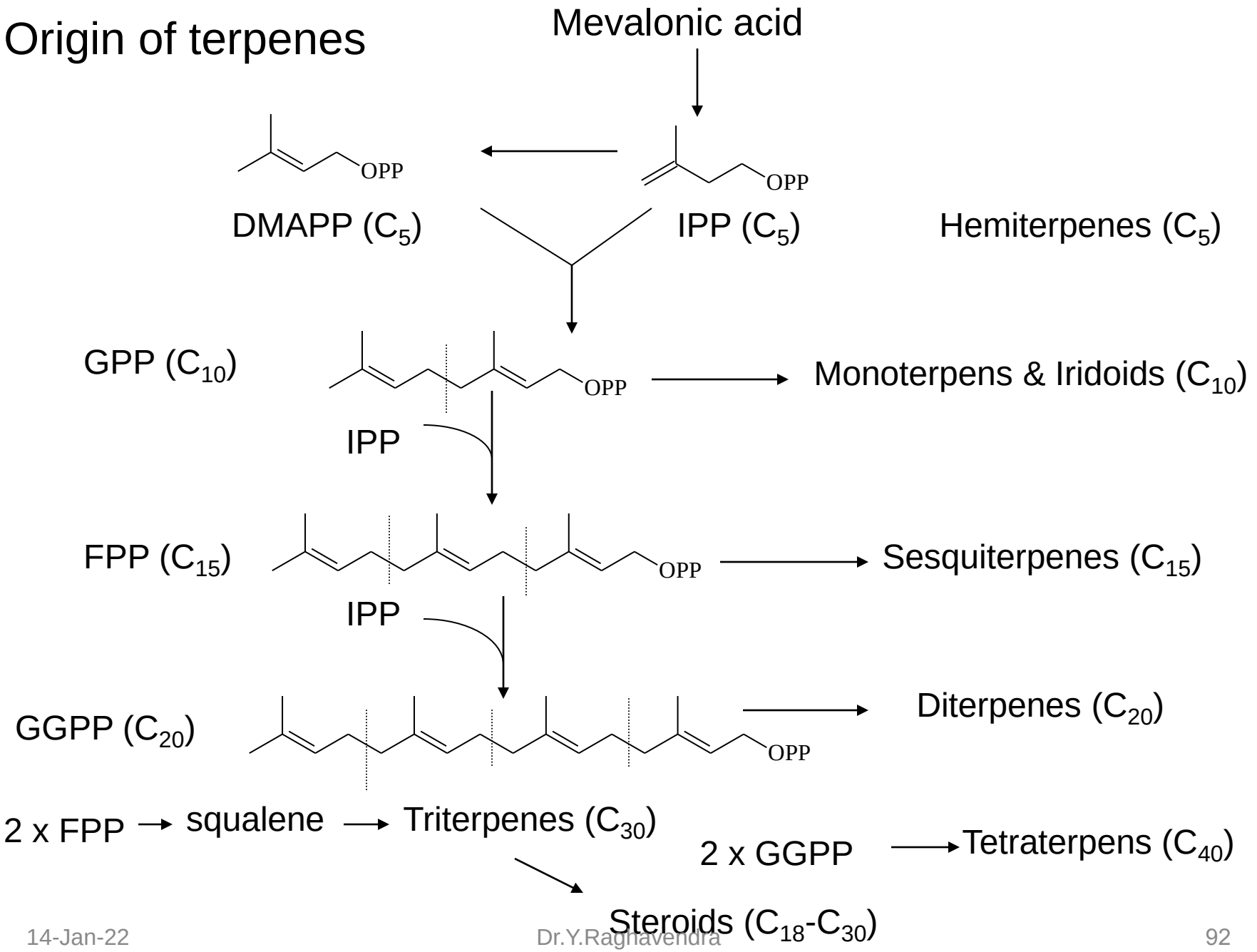
- Civet mainly contains civetone, civetol, indole, ethylamine, propylamine and few unidentified free acids.

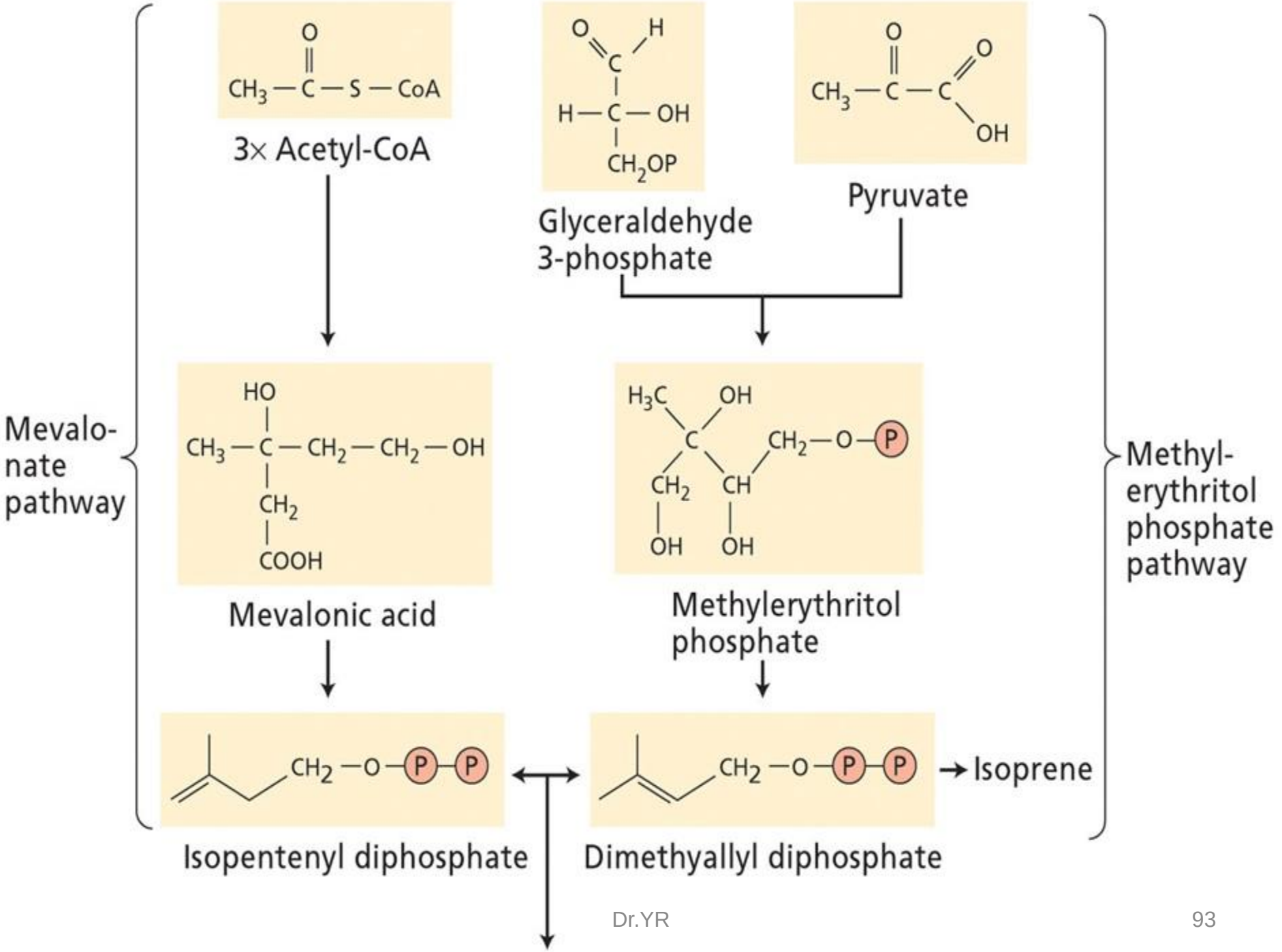
- It is used as a flavouring agent in cosmetics and foods
- As a fixative in perfumery.

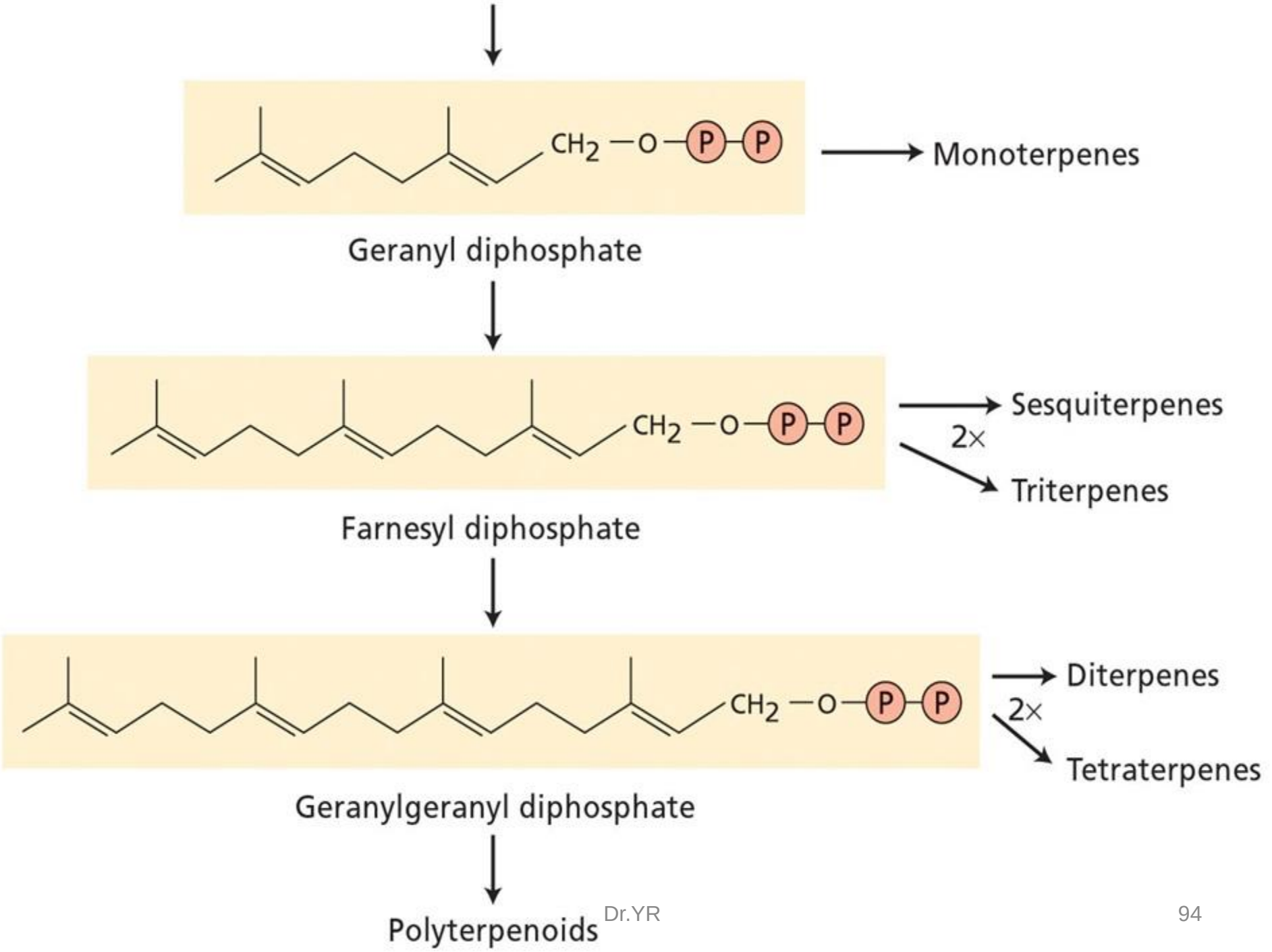
CHEMICAL COMPOSITION

- Volatile oils are divided into 2 main classes based on their biosynthetic origin
  - Terpene derivatives (formed via the acetate mevalonic acid pathway)
  - Aromatic compounds (formed via the shikimic acid-phenylpropanoid route)
  - Miscellaneous Origin

Origin of terpenes




3× Acetyl-CoA
Glyceraldehyde 3-phosphate
Pyruvate
Mevalo-nate pathway
Methyl-erythritol phosphate pathway
Mevalonic acid
Methylerythritol phosphate
Isopentenyl diphosphate
Dimethyallyl diphosphate
Isoprene


CH2 —O— P P
Monoterpenes
Geranyl diphosphate
CH2 —O— P P
Sesquiterpenes
2×
Triterpenes
Farnesyl diphosphate
CH2 —O— P P
Diterpenes
2×
Tetraterpenes
Geranylgeranyl diphosphate
Polyterpenoids

GINGER

- **Biological source:**
  - Ginger is the peeled or unpeeled rhizome of *Zingiber officinale*
- **Family:** Zingiberaceae
- **Geographical source:**
  - It is said to be native of South East asia, but is cultivated in carribean islands, Africa, Australia, mauritius, jamaica, Taiwan and India.
- Description:

Colour	-	Externally it is buff coloured
Odour	-	Agreeable and aromatic
Taste	-	agreeable and pungent

- Chemical constituents:
- Ginger consists of Volatile oils (1 – 4%)
  - Camphene, Cineole, Citral, Borneol
- Gingerol – pungent component – Anti-inflammatory
- Sesquiterpene HC's
  - Zingiberene & Zingiberol (Sesquiterpene alcohol)
- Resinous matter(5-8%), Starch (40-60%), Fat(10%), Fibre(5%).
- Uses:
  - It is used as a stomachic, an aromatic, a carminative, stimulant, Anti-emetic, Anti-bacterial & Anti-fungal and flavouring agent.
  - Ginger oil is used in mouth washes, beverages and liquors.
  - Ginger powder is effective in motion sickness.
  - Gingerol – pungent component – Anti-inflammatory
  - Shogaols– increases bile secretion & enhanced GIT activity.

CARDAMOM



- **Biological source:**
  - Cardamom consists of the dried, nearly dried ripe fruits of *Elettaria cardamomum* var *minuscula*
- **Family:** Zingiberaceae.
- **Geographical source:**
  - Guatemela is the largest producer in the world. Also cultivated in India and Srilanka.

➢ **Description:**

Colour	-	Natural colour is green or pale buff and after treated with sulphurdioxide) it changes to white.
Odour	-	Aromatic, agreeable and pleasant
Taste	-	Strongly aromatic

- Chemical constituents:
  - 2 – 8 % volatile oil.
  - The active constituent of the VO is cineole.
  - Other aromatic compounds present are terpinyl acetate, terpineol, borneol, terpinene etc.
  - Abundant starch (up to 50%) Fixed oil (1 – 10%), protein and Calcium oxalate.
- Uses:
  - It is used as an aromatic, as carminative and as stimulant.
  - It is also a good flavouring agent in preparation of curries, biscuits, sweets etc.
  - It is used in the form of compound tincture.

CINNAMON

- **Biological source:**
  - Cinnamon consists of the dried bark of the shoots of coppiced trees of *Cinnamomum zeylanicum*
- **Family:** Lauraceae
- **Geographical source**
- Cultivated in Sri Lanka, South India. Also cultivated in Seychelles, Madagascar, Martinique, Jamaica & Brazil.
- **Description:**

Colour	-	The outer surface is dull yellowish brown, while the inner surface is dark yellowish brown
Odour	-	fragrant
Taste	-	aromatic and sweet followed by ward sensation.
Shape	-	found in the form of compound quills.

- Chemical constituents:
  - 0.5 to 1.0% Volatile oils
  - Cinnamon oil contains 60-70% of cinnamaldehyde, 5-105 of eugenol, benzaldehyde, cuminaldehyde and other terpenes like phellandrene, pinene, cymene, caryophyllene, etc.



- The other constituents are 1.2% of tannins(Phlobaphenes), mucilage, Calcium Oxalate, Starch and a sweet substance called mannitol.

- Uses:
  - The bark is used as a carminative, stomachic and mild astringent.
  - It is also used as a flavouring agent, stimulant, an aromatic and antiseptic.
  - Commercially it is used as a spice and condiment and also in preparation of candies, dentrifices and perfumes.

LEMON PEEL

- **Biological source:**
  - Lemon peel is the outer part of the pericarp of the ripe or nearly ripe fruits of *Citrus limonis* Linn.
- **Family:** Rutaceae
- **Geographical source:**
  - It is indigenous to India, but also cultivated in countries like Spain, Italy and Sicily.
- **Description:**

Colour	-	(i) Fresh lemon peels outer surface is bright greenish yellow, internally white.
		(ii) Dried peels outer surface is yellow and inner surface pithy white.
Odour	-	Strong, fragrant, aromatic and characteristic
Taste	-	Aromatic, bitter.

- **Chemical constituents:**
  - Lemon peel contains 2 to 4% volatile oil.
  - The volatile oil contains mainly limonene (about 90%), citral(about 4%) and other aromatic compounds like geranyl acetate and terpineol.
  - The other constituents of the peel are Hesperidin (Flavonoid), pectin, Mucilage and Calcium Oxalate.

- **USES:**
  - It is used as carminative and stimulant and oil is used as a perfuming and flavouring agent.
  - It is used for extraction of pectin and volatile oil.

CARAWAY & CARAWAY OIL

- **Biological source:**
  - Caraway consists of the dried ripe fruits of *Caram carvi Linn.*
- **Family:** Umbelliferae
- **Geographical source:**
  - Indigenous to Holland Central & Northern Europe.
  - It is also cultivated in Central Asia and other european countries like Denmark, Germany, Finland, Poland, Hungary, Britain and also in India, Egypt, Morocco, Australia & China.
- Description:

Colour	-	Yellowish brown
Odour	-	characteristic and aromatic
Taste	-	characteristic and aromatic

- **Chemical constituents**:
  - Caraway consists of 2-8 % Volatile oils.
  - Volatile oil contains 45-65% of Carvone, Limonene, dihydro carvone and traces of carvacrol.
  - About 10 % Fixed oils, 15% Proteins and resins.
- **Uses**:
  - It is used as an aromatic, a stimulant and carminative.
  - It is used as a spice for culinary purposes and also for flavouring the bread, biscuits, cheese and cakes.

CORIANDER

- **Biological source:**
  - These are fully dried ripe fruits of plant known as *Coriandrum sativum* Linn.
- **Family:** Umbelliferae
- **Geographical source:**
  - Indigenous to Italy.
  - Also cultivated in Holland, Central & Eastern Europe, Mediterranean (Morocco, Malta & Egypt), China and India.
- **Description:**
  - Colour - Yellowish brown to brown
  - Odour - Aromatic
  - Taste - spicy and characteristic

- **Chemical constituents**:
  - 0.3 - 1.8 % Volatile oils
  - Volatile oil of drug contains about 90% of D-Linalool/coriandrol and coriandryl acetate.
  - Small quantities of L-borneol, geraniol and Pinene.
  - Coriander leaves are rich in Vitamin A content.
  - Non-linalool alchols & esters
  - Flavonoids, Coumarins, Phenolic acids,High fat content (16-28%), Protein (11-17%),
- **Uses**:
  - The fruits as well as volatile oil are used as an aromatic, carminative, stimulant and flavourign agent.
  - Coriander oil is used alongwith purgatives to prevent gripping.
  - It is an ingredient of compound spirit of orange and cascara elixir.
  - Domestic purposes (cooking - curries)

STORAGE OF VOLATILE OILS

- For all volatile oils, three conditions of storage are required, namely;
  - **1. Storage in Well-closed Containers,** To minimize:
    - Volatilization,
    - Exposure to air, and
    - Absorption of moisture

- **2. Protection of Oils from Light**
  - Volatile oils are protected from light by the use of amber-coloured bottles, well-stoppered and tightly closed container.
  - Light promotes many changes, particularly oxidation.
  - Further, it accelerates hydrolysis of esters if moisture is present.
  - It may also cause polymerization of the constituents of certain oils.
- **3. Storage in a Cool Place**
  - This requirement, ofcourse, reduces the possibility of loss by volatilization, and minimizes oxidation and other changes.

- Some examples of volatile oils effected due to improper storage
  - **Clove oil & Cinnamon oil:** These oils become reddish-brown upon exposure to air.
  - **Caraway oil:** This oil becomes thicker and its weight per ml. increases upon exposure to air.
  - **Anise oil:** This oil contains anethole, an alcohol, which upon exposure to air is slowly oxidized to anisic aldehyde, and this later converts to anisic acid.
  - **Dill oil:** This becomes darker on exposure.
  - **Bitter Almond oil:** Benzaldehyde, the principal active constituent of this oil, upon exposure to air undergoes oxidation to benzoic acid that separates as crystals.